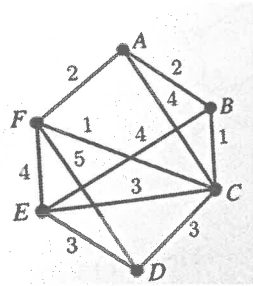
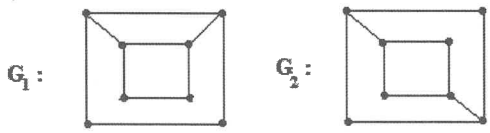
USN | | | | | | | | | | | [22MTAI11]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2024

MATHEMATICAL FOUNDATIONS FOR AI

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	I. A computer company required 30 programmers for handling system program jobs. 40 programmers for application programming job. If the company appoints 55 programmers to carry out these jobs, **i.** How many of these perform jobs of both type? **ii.** How many handles only system programming jobs? **iii.** How many handles only application programming job? II. Find the explicit form of the sequence defined by $a_n = na_{n-1}$ $with\ a_1 = 6$	**10**
	b.	Construct the truth table for the compound proposition $\{(p \vee q) \wedge [(p \rightarrow r) \wedge (q \rightarrow r)]\} \rightarrow r$	**10**
2.	**a.**	Using Prim's algorithm, find a minimal spanning tree of the weighted graph  	**10**
	b.	**I.** Show that the following graphs are isomorphic.  **II.** Given $A = \{1,2,3,4,6\}$ and R be a relation defined by xRy iff x is a multiple of y **i.** Write R as ordered pair. **ii.** Write M(R). **iii.** Draw directed graph of relation. Find indegree and out degree.	**10**
3.	**a.**	Test for consistency and hence solve the following system of equations $x + 2y + 2z = 1$ $2x + y + z = 2$ $3x + 2y + 2z = 3$ $y + z = 0$	**10**
	b.	Solve the system of equations by Gauss–Seidel method to obtain the final solution	**10**

		correct to three decimal places $x + y + 54z = 110;\ 27x + 6y - z = 85;\ 6x + 15y + 2z = 72.$	
		OR	
4.	**a.**	Three boys A, B, C are throwing ball to each other. A always throws the ball to B and B always throws the ball to C. C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball find the probabilities that after three throws **i.** A has the ball **ii.** B has the ball **iii.** C has the ball **iv.** Also find the associated unique fixed probability vector.	**10**
	b.	Solve the following system of equations by Gauss – Jordan method $2x + 5y + 7z = 52,\ \ 2x + y - z = 0,\ \ x + y + z = 9.$	**10**
5.	**a.**	Find all the Eigen values and the corresponding Eigen vectors of the matrix. $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	**10**
	b.	If $T: R^3 \to R^3$ defined by $T(x, y, z) = (x + y,\ x - y,\ 2x + z)$ then find R(T), N(T) and verify Rank – Nullity Theorem.	**10**
		OR	
6.	**a.**	Find A^{-1} if $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ using Cayley Hamilton theorem.	**10**
	b.	Show that the mapping $T: V_3(R) \to V_3(R)$ defined by $T(e_1) = e_1 + e_2,\ T(e_2) = e_2 + e_3$ and $T(e_3) = e_1 + e_2 + e_3$ is non – singular and find its inverse.	**10**
7.	**a.**	Examine the orthogonality of set $\{u_1,\ u_2,\ u_3\}$ where $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}\ \ u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}\ \ u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$.Express $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as linear combination of vectors.	**10**
	b.	Let W be the subspace of R^4 with basis $S = \{u_1,\ u_2,\ u_3\}$ where $u_1 = (1, -2, 0, 1)$, $u_2 = (-1,\ 0, 0,\ -1)$ and $u_3 = (1, 1, 0, 0)$. Use the Gram- Schmidt process to transform S to orthonormal basis for W.	**10**
		OR	
8.	**a.**	The distance from a point v in R^n to a subspace w is defined as the distance from v to the nearest point in w. Find distance from v to w ($w = span\{u_1, u_2\}$) Where $v = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix}$, $u_1 = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$.	**10**
	b.	Consider the basis $S = \{u_1,\ u_2,\ u_3\}$ forR^3 where $u_1 = (1,1,1)$, $u_2 = (-1,\ 0,\ -1)$ and $u_3 = (-1,\ 2,\ 3)$. Use the Gram- Schmidt process to transform S to orthonormal basis for R^3.	**10**